

Yash Swaminathan

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Education

University of Waterloo

Bachelor of Applied Science (BASc) in Systems Design Engineering (Co-op)

Waterloo, ON

Sep. 2024 – Present

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL, Java, C++, HTML/CSS

Infrastructure: Docker, AWS (S3, SQS, EC2), Terraform, GitHub Actions, Kafka, k6, MinIO, Git, Linux

Observability: Prometheus, Grafana, Grafana Tempo, New Relic, OpenTelemetry

Frameworks & Databases: FastAPI, Node.js, Express.js, PostgreSQL, Redis, Spring Boot, React, Next.js, ScyllaDB, TiDB

Libraries: Stripe SDK, NumPy, Pandas, TensorFlow, Pytest, Jest, Pydantic, PyYAML, jsonschema, Socket.io

Experience

Software Engineer Intern — Backend & Infrastructure

May 2026 - Present

StackAdapt

Toronto, ON

- Instrumented **22 Go microservices** across the 1P data ingestion pipeline using **OpenTelemetry**, adding spans and trace context propagation across **Kafka** async boundaries, enriching spans with missing **CDP** event-type and profile attributes.
- Scaled distributed tracing to **860K+ sampled traces/day (8.6M unsampled)** across **100+ events/sec** of CDP ingestion data, storing **3–5 GB/day** of compressed trace data in **Grafana Tempo** via AWS S3 for end-to-end pipeline visibility.
- Normalized **880K+ records/day** from 1,000+ Shopify shops alongside **Salesforce, Snowflake, and HubSpot** sources by tracking each source's ingestion ID per period, producing one consistent, ingestible schema across all platforms.

Software Engineer Intern — Backend & Infrastructure

Sep 2025 – Dec 2025

Micromart

Toronto, ON

- Modelled infrastructure capacity for **10,000 machines** through **performance analysis and cost modelling**, enabling fleet expansion from **900 → 2,100 machines in one month** and deployment of **\$7.2M in hardware**.
- Designed **k6 load tests** on TypeScript/Node.js backend simulating **20,000 RPS** and **15,000 concurrent WebSocket connections**, targeting API calls consuming **95% of total backend request time** to identify scalability bottlenecks.
- Built observability pipelines in **New Relic and Grafana** with distributed tracing and custom dashboards tracking **API latency, resource utilization, cron jobs, and service health**, reducing root cause identification time by **60%**.
- Optimized backend performance through **Redis caching**, refactoring system heartbeat endpoints, and **retry/backoff logic** on machine restart/shutdown, reducing **P95 latency by 35%** and total **API call time by 15%**.

Software Engineer Intern — Backend & DevOps

Jan 2025 – Apr 2025

Turing

Calgary, AB

- Designed and implemented **Alembic Migration** scripts for **PostgreSQL** test databases, automating test database creation and configuration, reducing manual setup time by **200+ hours**, resulting in **\$15,000+** of savings.
- Developed **10+ RESTful API endpoints (FastAPI)** and authored **60+** unit tests to achieve **95%** coverage, ensuring reliable API behaviour and maintenance of critical features.

Data Science Intern

Sep 2023 – Jan 2024

Gradiant (Synauta)

Calgary, AB

- Developed a **Python-based analytics pipeline** to process **IoT sensor data**, identifying optimal cartridge filter replacement timing and reducing premature replacements by **15–20%**.

Projects

ChatterBox - Real-Time Messaging Platform | *Node.js, Express, PostgreSQL, Redis, Socket.io, Docker, MinIO, Jest, k6*

- Built real-time messaging backend with **Redis pub/sub** enabling horizontal scaling across multiple servers, supporting **1000+ concurrent WebSocket connections** through efficient room-based message broadcasting.
- Engineered **group messaging system** with admin roles, participant management, and PostgreSQL full-text search with @mention parsing, using advisory locks to prevent race conditions in groups with **100+** members.
- Implemented **JWT-based authentication** with **bcrypt password hashing**, refresh token rotation, and device-based session management for secure multi-device access.

E-Commerce Platform | *Java, Spring Boot, Go, TypeScript, React (Next.js), Express.js, Prisma, Stripe SDK*

- Designed event-driven microservices architecture (**Go + Spring Boot**) that reduced order-processing latency from **800ms to 480ms**, enabling real-time inventory updates and preventing overselling across **10+ concurrent orders**.
- Implemented secure payment flow with **Stripe webhooks** for async order confirmation and **AWS S3** for product media, processing **1,000+ test transactions** with automated payment reconciliation.
- Automated deployment pipeline with **GitHub Actions** and **Terraform**, achieving **60% faster releases** (25min → 10min) with zero-downtime blue-green deployments and automated rollback capabilities.